# YUGENDHRA E

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## SUMMARY

Information Technology graduate and Software Developer passionate about backend systems, distributed architectures, and scalable full-stack engineering. Experienced in architecting REST APIs, asynchronous task queues, and real-time event-driven applications using TypeScript, Node.js, Express, React, MongoDB, Redis, and Docker.

## EDUCATION

**B.Tech, Information Technology** Nov 2022 – May 2026

Anna University, Chennai, Tamil Nadu CGPA: 8.77

Agni College of Technology

Relevant coursework: Data Structures & Algorithms, Operating Systems, Computer Networks, DBMS, OOP Concepts

## PROJECTS

**Sotrix – Full-Stack Social Networking Platform** Live Demo | GitHub

Built a full-stack real-time social platform using TypeScript, React 19, Node.js, Express, MongoDB, and Redis.

- Built a full-stack social platform with 45+ REST endpoints across 10 modules, real-time messaging, and custom timeline feeds
- Secured the platform against brute-force attacks and unauthorized access using Zod schema validation, tiered rate limiting, and dual-token JWT auth
- Implemented background worker queues with BullMQ and Redis for Sharp image compression, cursor-based feed pagination, and Socket.IO with JWT handshakes

**Cappd – Distributed Resource Budgeting Engine** npm | GitHub

Built an open-source distributed capacity reservation engine for variable-cost workloads using TypeScript, Node.js, and Redis.

- Developed a resource budgeting engine for backend systems to manage variable-cost operations like LLM tokens, compute jobs, and user credit limits
- Solved the limitation of standard rate limiters that only count requests, preventing high-cost operations from exceeding quotas and eliminating concurrency race conditions
- Implemented atomic two-phase holds with Redis Lua scripts to reserve estimated usage upfront and refund unused balance; built Express middleware and published to npm with 98 Vitest tests

**SpendWise – Expense Tracker REST API** Live Demo | GitHub

Built a secure RESTful backend for personal expense management using Node.js, Express.js, MongoDB, and JWT.

- Designed REST APIs with JWT-based authentication and bcrypt password hashing
- Implemented expense CRUD operations, pagination, category-based filtering, and user-specific data access
- Containerized the application using Docker for consistent development and simplified deployment

## EXPERIENCE

**Altruisty**, Full Stack Development Intern Oct 2024 – Dec 2024

- Designed the front-end UI for a College Management System focusing on usability and layout consistency
- Gained hands-on experience in UI structuring and front-end design workflows

**SystemTron**, Web Development Intern Mar 2024 – Apr 2024

- Developed a Netflix-inspired static website as the primary internship project
- Built mini projects including a Calculator and Connect 4 Dots game to strengthen front-end fundamentals

## SKILLS

**Languages:** C++, JavaScript, TypeScript, Python, Java, C

**Frontend:** React, Next.js, HTML, CSS, Zustand, TanStack Query

**Backend:** Node.js, Express.js, REST APIs, Socket.IO, BullMQ, Zod, WebRTC

**Databases:** MongoDB, PostgreSQL, MySQL, Redis, Firebase, Supabase

**Tools:** Git, GitHub, Docker, Postman, Swagger, Jest, Render, Vercel, VS Code

## ACHIEVEMENTS

**Academic Excellence Award**, Agni College of Technology 2022 – 2023

- Received the award for securing a GPA of 9.0 in the university's 1st semester examination

## ACTIVITIES

**Iris Club**, President 2024 – 2025

- Led a team of 25+ members to manage end-to-end technical event planning, workshops, and team coordination